

P KAVIN

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Education

Indian Institute of Technology Kharagpur

5-Year Dual Degree (BS Chemistry + M.Tech AI and ML Applications)

Aug. 2023 – Present

CGPA : 8.17/10

Coursework and MOOCs

- AI Foundations and Applications
- Foundations of LLMs
- Sequence Models
- Linear Algebra for AI and ML
- Stochastic Processes and Applications
- Attention Models
- Statistical Foundations of AI/ML
- Probability and Statistics

Experience

Applied AI Lab, UNIL HEC Lausanne — Summer Research Intern

May. 2026 – Jul. 2026

Supervisor: Prof. Yash Raj Shrestha

Lausanne, Switzerland

- Benchmarked several multilingual ASR and speech translation models - Whisper, SeamlessM4T, Gemma 4, TranslateGemma, OmniASR, Qwen3, etc. across 20 different languages using evaluation pipelines.
- Developed an end-to-end framework for multilingual speech systems using CER, WER, MER, WIL, BLEU, and COMET, enabling comparison of transcription accuracy and translation quality across diverse languages
- Conducted large-scale model inferencing on the University of Lausanne DCSR cluster to evaluate ASR and speech translation models.
- Evaluated multilingual speech recognition and translation for low-resource languages - Somali, Tigrinya, Georgian, Kurdish, Romansh, etc

Indian Institute of Technology Bombay — Research Intern

Nov. 2025 – Present

Supervisor: Prof. Preeti Rao

Remote

- Evaluated end-to-end robustness of Tamil ASR models on a speech dataset from 60 school students, simulating practical classroom environment.
- Benchmarked IndicWav2Vec, Indic Conformer CTC, and Indic Conformer RNN-T models for Tamil Automatic Speech Recognition using Character Error Rate (CER) to compare transcription performance.
- Generated verbatim Tamil transcripts by manually listening to and annotating speech recordings, creating high-quality ground-truth data for ASR model evaluation.
- Assessed noise robustness by augmenting classroom recordings with realistic background noises - fan, traffic, and aircraft at 5 dB, 10 dB, 15 dB, and 20 dB SNR, analyzing the impact of acoustic conditions on ASR accuracy.

Speech and Language Lab, NTU Singapore — Speech Intern

Jul. 2026 – Present

Supervisor: Prof. Chng Eng Siong

Remote

- Developing 1,000 hours of Tamil speech data from online sources to build a large-scale corpus targeting low-resource Automatic Speech Recognition (ASR) research.
- Adapting a multilingual ASR data-crawling framework for Tamil, implementing subtitle filtering and timestamp-based slicing of YouTube audio.

Projects

Transformer-based Neural Machine Translation for English to Tamil | Self Project

Jul. 2025 - Sep. 2025

- Implemented a miniature Transformer architecture from “Attention is All You Need” for Tamil language translation.
- Trained on a parallel English–Tamil corpus of 1.18 Lakh text pairs using Encoder and Decoder Blocks.
- Finetuned hyperparameters of the model to minimise the Cross Entropy, achieving a lowest value of 3.3304.
- Developed the model using PyTorch and optimised training for convergence.

Water Solubility Prediction of Organic molecules using Machine Learning | Self Project

Mar. 2025

- Developed a Machine Learning predictive model to estimate the solubility of organic compounds.
- Performed feature engineering on the dataset, to bridge the gap between chemistry and data science.
- Achieved the highest accuracy with an R2 score of 0.92 using XGBoost and 0.89 using Random Forest.
- Optimized model performance through hyper-parameter tuning, significantly reducing the outliers.

Brain Tumour Image Segmentation using Deep Learning | *Self Project*

Jan. 2025

- Constructed a U-Net model in TensorFlow/Keras for MRI-based Brain Tumour Segmentation.
- Utilised a dataset of 3000 labelled images for supervised semantic segmentation.
- Evaluated model performance using Binary Cross-Entropy Loss and achieved an Intersection over Union(IoU) of 0.71.
- Enhanced segmentation accuracy through extensive data augmentation and learning rate tuning.

Multi-class Image Classification of Galaxies using Convolutional Neural Networks | *Self Project*

May. 2025

- Built a Computer Vision model to classify galaxies of 6000 images with 10 classes from Galaxies10 Decals dataset.
- Developed models from scratch using TensorFlow/Keras with an accuracy of 52%.
- Pre-Trained ResNet50 and EfficientNetB3 on the test dataset.
- Reached 74.1% accuracy with ResNet50 and 75.2% with EfficientNetB3.

Asset Price Prediction using RNNs | *Finance and Economics Club IIT Kharagpur*

Jun. 2025

- Designed LSTM, BiLSTM, and GRU networks to forecast short-term stock prices.
- Visualised financial trends through moving averages and comparative prediction plots.
- Achieved Mean Absolute Errors of 0.0211 (train) and 0.0222 (test).
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Technical Skills

Languages: Python, C

Developer Tools: VS Code, Google Colab, Kaggle, Jupyter, HuggingFace.

Technologies/Frameworks: GitHub, TensorFlow, PyTorch, Scikit-Learn, Matplotlib, Seaborn, Pandas, Jiwer.

Extracurricular

- Accomplished Grade 8 in Drums from Trinity College London in 2022.
- Accomplished Grade 2 in Guitar from Trinity College London in 2021.
- Selected for NCA-Instrumental, being one among 30 students chosen.
- Pursued training in Harmonium under the Academy of Classical and Folk Arts, IIT Kharagpur.